# Coherent systems

- IM/DD vs Coherent Detection (CD)
- Block diagram of CD
- Coherent Detection, CD Gain
- Practical constraints
- Coherent systems - Modulation formats
- SNR & shot noise limit
- Multicarrier system
- Implementation of Coherent Detection for laser linewidth measurements

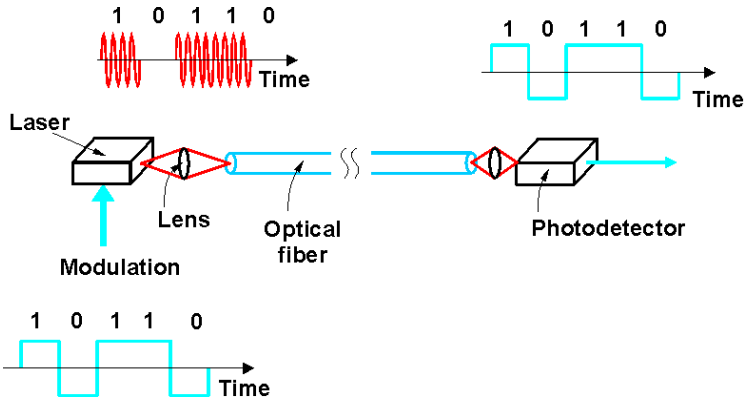
## Intensity Modulation/Direct Detection IM/DD Systems
## vs
## Coherent Systems (Coherent Detection CD)

Adventages of coherent systems over IM/DD:

- Higher sensitivity (10-20 dB):
  IM/DD requires 700 – 20.000 photons/bit for BER=$10^{-9}$
  CD: 20-100 ph/bit at BER=$10^{-9}$
- Increased selectivity
  (channel selection in electrical domain)

# IM/DD – ignores the wave nature of electromagnetic carrier

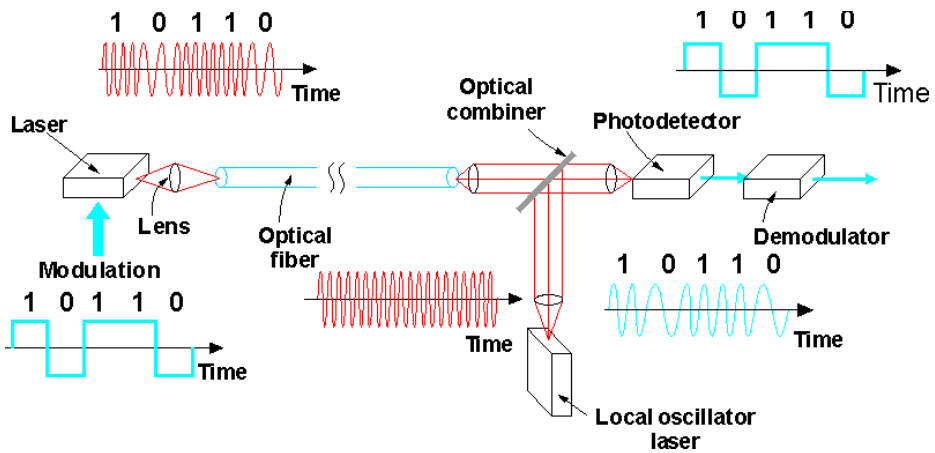
Direct detection is a photon counting, optical power measuring process



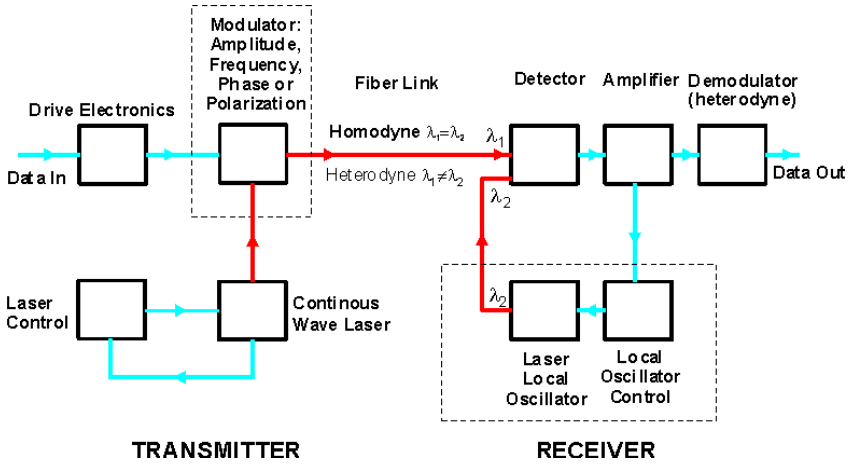
# Intensity Modulation/Direct Detection IM/DD Systems

Conventional direct detection receivers are limited by noise generated in the detector and preamplifier

Coherent system takes into account wave nature/parameters of optical beam: phase, frequency and amplitude



# Coherent system




# Coherent systems schemes

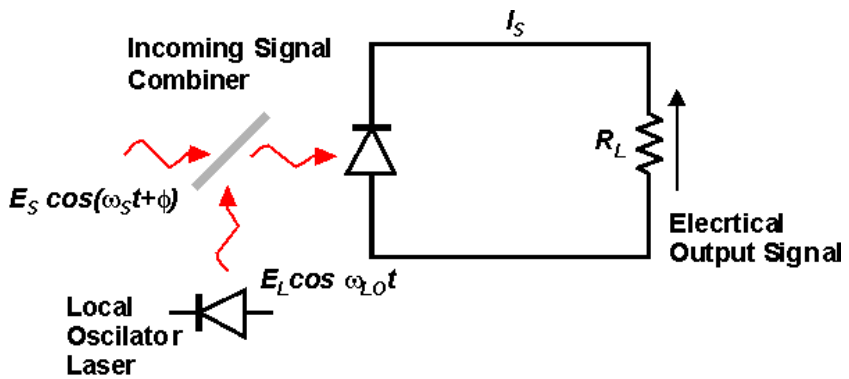
| DETECTION/ MODULATION SCHEME | HOMODYNE | HETERODYNE SYNCHRO. | HETERODYNE ASYNCHR. |
|---|---|---|---|
| ASK | X | X | X |
| FSK | X | X | X |
| PSK | X | X | X |
| PolSK | X | X | X |
| | | | |

# Comparison of theoretical and experimental results

| CD | IM/DD |
|---|---|
| 9-40 fot/bit | 10,4 fot/bit |
| 2 dB | 10-20 dB |

**JEŻELI JEST TAK DOBRZE (10,4 ppb)**
**TO DLACZEGO**
**JEST TAK ŹLE (20 dB)**
**?!**
**If it is so good why is it so bad ?**

# Coherent detection



$$e_s = A_s \cos\left(\omega_s t + \varphi_s\right)$$
$$e_{LO} = A_{LO} \cos\left(\omega_{LO} t + \varphi_{LO}\right)$$

$$P_t = \left|A_s + A_{LO}\right|^2 = P_s + P_{LO} + 2\sqrt{P_s P_{LO}} \cos\left(\left|\omega_s - \omega_{LO}\right| t + \varphi\right)$$

$$I_p = RP_t = R\left(P_s + P_{LO}\right) + 2R\sqrt{P_s P_{LO}} \cos\left(\left|\omega_s - \omega_{LO}\right| t + \varphi\right)$$

## Coherent detection

$$R\left(P_s + P_{LO}\right) \approx const$$

$$I_s = \frac{2\eta e}{hf} \sqrt{P_s P_{LO}} \cos\left(\omega_{IF} t + \varphi\right)$$

## Gain of coherent system:

$$G = \frac{I_s^{CD}}{I_s^{DD}}$$

$$I_s^{CD} = 2R\sqrt{P_s P_{LO}}$$

$$I_s^{DD} = RP_s$$

$$G = \frac{R\sqrt{P_s P_{LO}}}{RP_s} = \sqrt{\frac{P_{LO}}{P_s}}$$

## Practical constraints

- Laser linewidth:

  LEC, DFB, DBR lasers

- State of polarization:

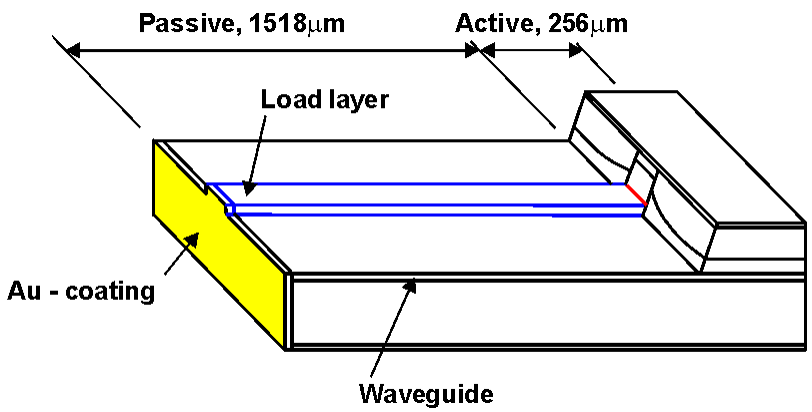
  Polarization maintaining fiber
  Polarization state controller
  Polarization diversity receiver
  Polarization scrambling transmitter

- Local oscillator power

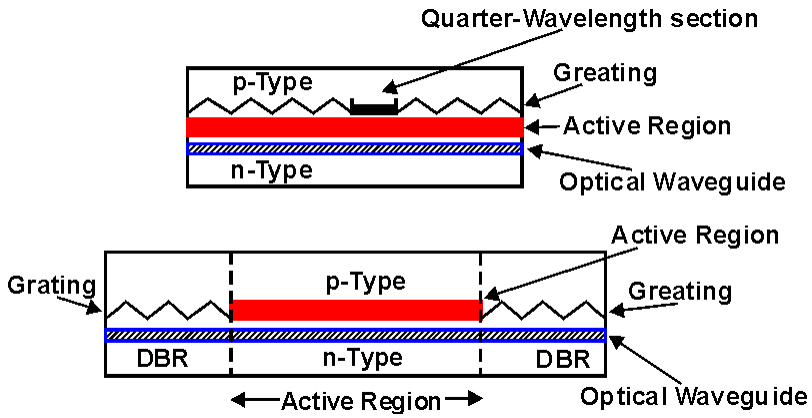
# LD spectral requirements

| MODULATION SCHEME | HOMODYNE | HETERODYNE SYNCHRO. | HETERODYNE ASYNCHR. |
|---|---|---|---|
| ASK | 0.005 - 0.1% | 0.05 - 0.1% | 10 - 50% |
| FSK (duża dew.) | | 0.05 - 0.1% | 10 - 50% |
| FSK | | | 0.3 - 2.0% |
| PSK | 0.005 - 0.01% | 0.1 - 0.5% | |
| DPSK | | 0.3 - 0.5% | |

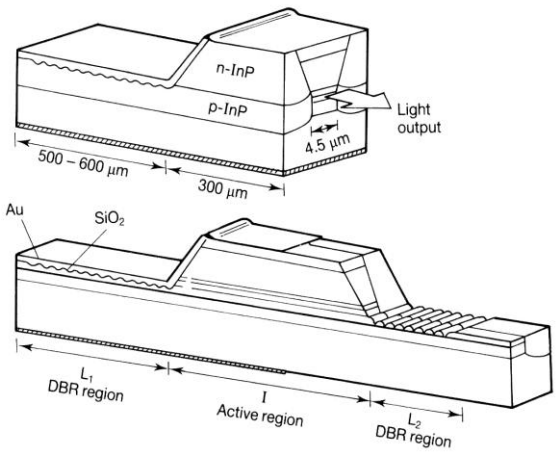
# LEC



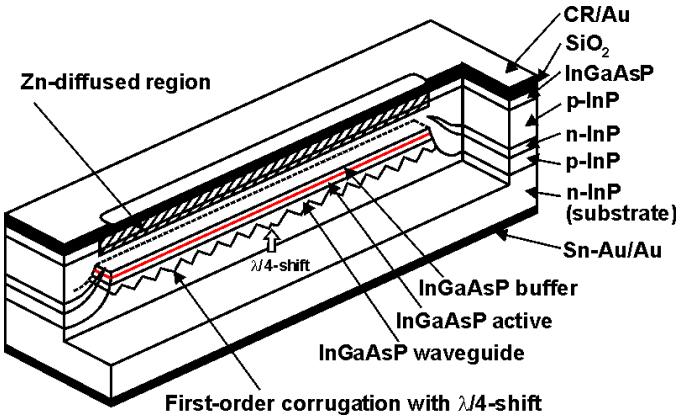
# DBR / DFB LD



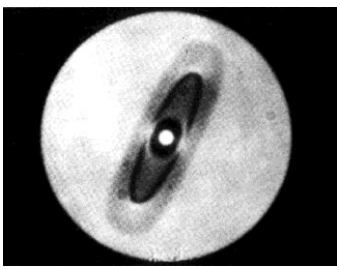

# DBR lasers



# DFB lasers

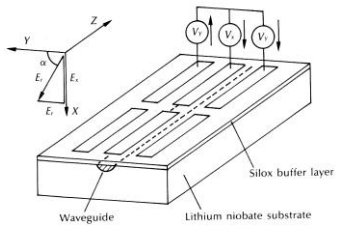



# PM Polarization maintaining fibers



# Polarization state control





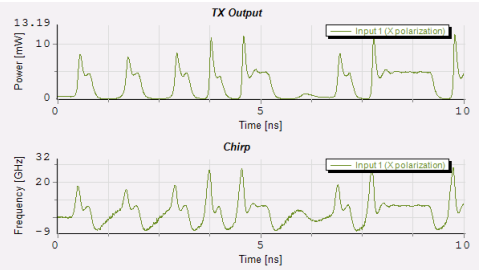

# Coherent systems - Modulation formats:

- Amplitude shift keying
  - Direct intensity modulation (unwanted effect: frequency chirp)
  - External modulation
- Frequency shift keying
  - Intensity modulation = frequency modulation
  - Acousto optic modulation
  - E/O modulation
- Phase shift keying
- Polarization shift keying

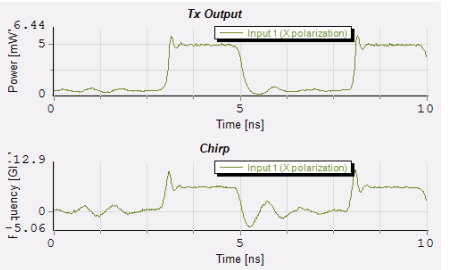
# Bit rate limitation due to Dispersion

- The waveform and chirp for 1 Gbit/s and 2.5 Gbit/s signal are shown below.

**Waveform and chirp of 2.5Gbit/s signal**

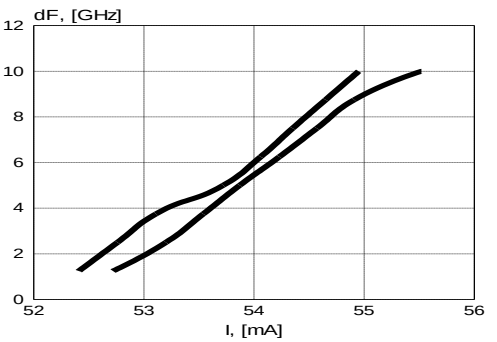


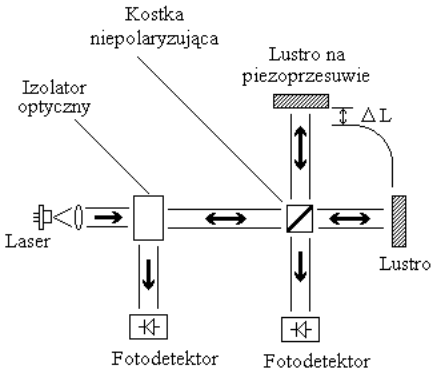

**Waveform and chirp of 1 Gbit/s signal**



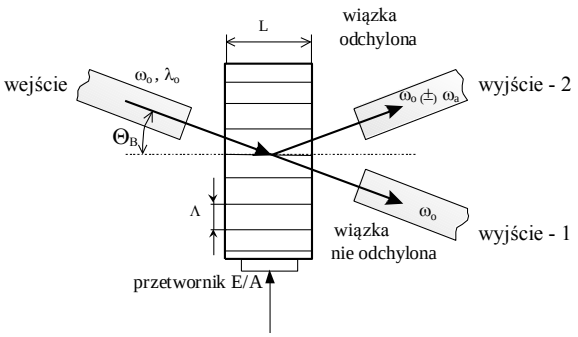

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# FM






# A-O Bragg Cell



# LD spectral requirements

| MODULATION SCHEME | HOMODYNe | HETERODYNe SYNCHRo. | HETERODYNe ASYNCHR. |
|---|---|---|---|
| ASK | 0.005 - 0.1% | 0.05 - 0.1% | 10 - 50% |
| FSK (duża dew.) | | 0.05 - 0.1% | 10 - 50% |
| FSK | | | 0.3 - 2.0% |
| PSK | 0.005 - 0.01% | 0.1 - 0.5% | |
| DPSK | | 0.3 - 0.5% | |

# SNR shot noise limit

$$I_p = \frac{\eta e}{hf} P_0 = \frac{\eta e \lambda}{hc} P_0$$

$$\overline{i_{shot}^2} = 2eBI_p \qquad \overline{i_{shot}^2} = \frac{2e^2 \eta P_{LO} B}{hf} = N$$

$$i_s = R\sqrt{P_S P_{LO}}$$

$$S = \left(\frac{\eta e}{hf}\right)^2 P_S P_{LO} \qquad \frac{S}{N} = \frac{\left(\frac{\eta e}{hf}\right)^2 P_S P_{LO}}{\frac{2e^2 \eta P_L B}{hf}} = \frac{\eta P_S}{hfB}$$

# SNR shot noise limit

$$\frac{\eta P}{hf} = r$$ Incident photon rate

$$\frac{\eta P}{hfB} = \frac{r}{B} = N_p$$ Number of photons per bit

$$\frac{S}{N} = \frac{4R^2 P_S P_{LO}}{\left(2eRP_{LO} + \frac{4kT}{R_L}\right)B}$$

# Receiver sensitivities in quantum limit

| Receiver: <br> Modulation: | Homodyne detection | Heterodyne Synchr. Det. | Heterodyne Asynchr. Det. | Direct Detection |
|---|---|---|---|---|
| ASK / OOK <br> Av. $N_p$pb <br> (two bit period) | **18** <br> $\frac{1}{2} erfc\left(\frac{\eta P_s}{2hfB_T}\right)^{0.5}$ | **36** <br> $\frac{1}{2} erfc\left(\frac{\eta P_s}{4hfB_T}\right)^{0.5}$ | **40** <br> $\frac{1}{2}\exp-\left(\frac{\eta P_s}{4hfB_T}\right)$ | **10,4** <br> $\frac{1}{2}\exp-\left(\frac{\eta P_s}{hfB_T}\right)$ |
| FSK <br> Av. $N_p$pb | | 36 <br> $\frac{1}{2} erfc\left(\frac{\eta P_s}{2hfB_T}\right)^{0.5}$ | 40 <br> $\frac{1}{2}\exp-\left(\frac{\eta P_s}{2hfB_T}\right)$ | |
| PSK <br> Av. $N_p$pb | **9** <br> $\frac{1}{2} erfc\left(\frac{2\eta P_s}{hfB_T}\right)^{0.5}$ | **18** <br> $\frac{1}{2} erfc\left(\frac{\eta P_s}{hfB_T}\right)^{0.5}$ | **20** DPSK <br> $\frac{1}{2}\exp-\left(\frac{\eta P_s}{hfB_T}\right)$ | |

# IM/DD practical sensitivity

| | $N_{p.}$/bit | Syst. Degrad. |
|---|---|---|
| Theor. Limit | 10 | 0 |
| Best results | | |
| InGaAs pin PD | 6.000-20.000 | 28-33 dB |
| InGaAs APD | 500-2.000 | 17-23 dB |
| EDFA+InGaAs | 147 | 11,7 |

# Multicarrier system

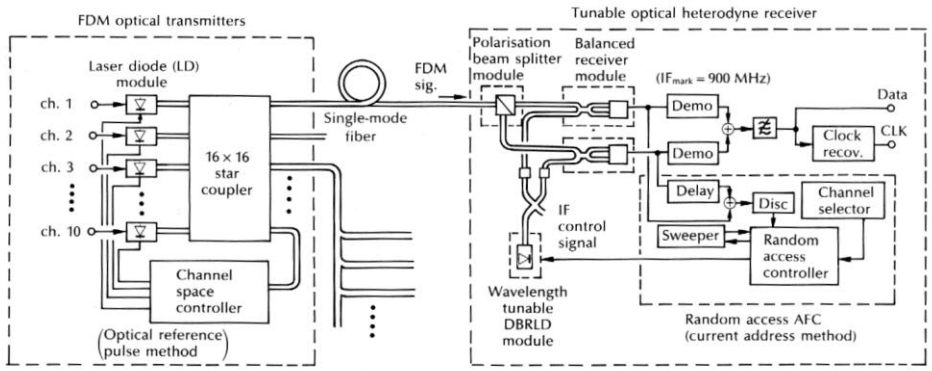



**Figure 12.29** Experimental ten channel coherent optical frequency division multiplexed CATV system [Ref. 108].

# Multicarrier system

$$P_s \geq N_P hfB_T$$ Power required at receivers

$$P_{tx} \geq N_P hfB_T N$$ Power required from transmiters

$$B_T \leq \frac{B_{fib}}{N}$$ Terminal transmission capacity

$$P_{tx} \geq N_P hfB_{fib}$$ Power required from transmiters

FDM, $N_p$=150 p/bit, $B_{fib}$=20 THz, $\lambda$=1.3 $\mu$m
$P_{tx}$=?

$$P_{tx} = N_p hfB_{fib} = \frac{N_p hcB_{fib}}{\lambda} =$$
$$= \frac{150 \times 6.63 \times 10^{-34} \times 3 \times 10^8 \times 20 \times 10^{12}}{1.3 \times 10^{-6}} =$$
$$= 0.5[mW] = -3[dBm]$$

